

HUMAIN, Belgium

WMO: 064720

Lat: 50.1936N

Lon: 5.2556E

Elev: 971

StdP: 14.19

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	21.9	10.9	10.0	22.2	17.4	13.7	25.0	29.0	45.2	26.2	45.4	7.4	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	84.9	67.8	80.4	66.2	76.9	64.5	70.1	80.3	68.0	76.6	66.1	74.3	7.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	102.4	73.7	64.9	95.5	71.8	63.0	89.4	69.5	34.6	80.5	32.8	76.3	31.3	74.6	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 22.6	19.5	17.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	13.4	73.5	4.4	2.6	10.2	75.4	7.6	76.9	5.2	78.4	1.9	80.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	35.6	36.3	43.2	48.8	53.9	61.0	64.7	63.0	57.7	51.7	43.2	37.7		
	DBStd	12.10	6.99	8.12	7.30	7.11	7.17	6.19	6.12	5.56	5.53	6.91	6.23	6.69		
	HDD50	1869	446	385	228	107	37	0	0	0	3	62	218	383		
	HDD65	5717	911	804	676	486	352	155	80	106	231	414	655	847		
	CDD50	1796	0	2	17	71	159	329	455	402	233	114	13	1		
	CDD65	165	0	0	0	0	8	34	70	42	10	1	0	0		
	CDH74	1476	0	0	0	5	67	314	626	387	76	1	0	0		
	CDH80	409	0	0	0	0	7	75	193	124	10	0	0	0		
	WSAvg	8.2	9.9	9.1	9.4	8.1	7.0	7.3	6.8	6.9	7.0	8.3	8.7	9.7		
	PrecAvg	40.9	4.2	3.4	3.1	2.4	2.9	3.1	3.5	3.8	3.0	3.3	3.5	4.6		
	PrecMax	50.8	10.6	8.2	6.7	6.3	6.7	6.3	8.3	9.5	6.5	7.7	5.9	10.9		
(15) Precipitation	PrecMin	30.0	0.5	0.5	0.4	0.2	1.2	0.8	1.7	0.9	0.5	1.4	0.2	0.7		
	PrecStd	5.4	2.0	1.7	1.7	1.5	1.4	1.5	1.5	2.2	1.8	1.3	1.5	2.2		
	0.4%	DB	52.7	60.3	66.5	74.0	80.4	86.7	91.6	89.1	81.0	73.0	60.2	53.8		
	MCWB	51.1	45.9	54.3	58.4	65.4	67.9	68.1	69.1	67.2	61.8	55.6	50.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	50.2	51.4	62.6	70.6	75.1	81.3	85.2	83.4	75.8	68.7	57.1	51.4		
	MCWB	49.0	45.5	51.5	57.0	61.7	67.8	68.4	67.0	64.7	60.6	54.4	49.3			
	5%	DB	47.3	48.3	58.4	66.0	71.1	77.3	80.5	77.8	71.7	64.5	54.9	48.9		
	MCWB	46.2	45.6	49.9	55.5	60.2	64.5	65.9	65.2	62.4	59.0	53.0	47.6			
	10%	DB	45.1	46.2	54.2	61.4	67.0	73.3	76.6	73.5	67.7	61.3	52.1	46.4		
	MCWB	43.9	44.2	48.2	53.0	58.5	63.2	65.0	64.0	60.7	57.1	50.3	45.4			
	0.4%	WB	52.0	49.3	54.9	60.7	68.2	72.3	72.0	72.5	69.0	64.1	56.8	52.1		
	MCDB	52.2	53.0	63.8	70.8	77.7	81.9	82.9	84.3	76.8	70.1	58.6	52.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.1	47.5	53.1	58.3	64.2	69.3	69.8	69.0	66.2	61.7	54.9	49.4		
	MCDB	49.9	49.8	60.1	67.9	71.8	78.4	81.1	78.0	74.0	66.3	56.6	50.7			
	5%	WB	46.4	45.7	50.8	55.9	61.6	66.2	67.8	66.4	63.7	59.6	53.0	47.7		
	MCDB	47.0	48.0	56.7	64.0	68.1	74.0	77.3	74.0	69.9	63.2	54.8	48.9			
	10%	WB	44.1	44.1	48.8	53.7	58.9	64.0	66.0	64.6	61.5	57.9	50.5	45.5		
	MCDB	45.0	46.4	53.5	60.8	65.5	71.7	74.9	72.1	66.7	60.9	52.0	46.5			
	5% DB	MDBR	7.0	9.3	13.8	16.7	15.9	16.6	17.1	16.2	15.5	12.1	8.3	7.5		
	MCDBR	8.7	12.9	22.3	22.9	22.1	23.2	23.4	23.9	21.0	16.8	11.0	10.0			
(36) Mean Daily Temperature Range	MCWBR	8.0	9.4	13.3	13.1	12.0	11.6	10.4	11.1	11.6	10.1	8.8	8.7			
	5% WB	MCDDBR	8.2	11.3	18.4	20.5	19.1	20.2	20.0	19.1	18.6	14.5	10.1	9.2		
	MCWBR	7.9	9.0	12.0	12.3	11.5	11.1	10.1	10.5	11.2	9.6	8.6	8.5			
(40) Clear-Sky Solar Irradiance	taub	0.306	0.330	0.362	0.393	0.403	0.403	0.402	0.396	0.366	0.349	0.323	0.304			
	taud	2.440	2.394	2.321	2.256	2.264	2.284	2.280	2.324	2.388	2.436	2.443	2.414			
	Ebn at Noon	234	256	266	269	271	272	269	265	262	245	224	215			
	Edh at Noon	19	26	33	39	41	40	40	36	31	25	19	17			
(44) All-Sky Solar Radiation	RadAvg	251	470	841	1323	1567	1720	1636	1394	1059	620	294	200			
	RadStd	38	90	118	158	165	158	184	153	112	53	35	33			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page